

PE-EC603C — CMOS VLSI Design

Semester VI • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: MOSFET Current Derivation & Scaling

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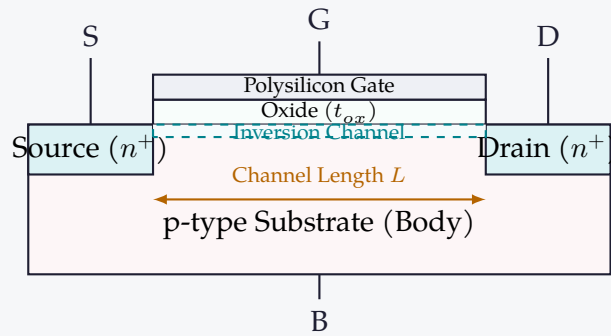
1. MOSFET Current Expression Derivation

To derive the current-voltage ($I_D - V_{DS}$) characteristics of a MOSFET, we utilize the **Gradual Channel Approximation (GCA)**.

Gradual Channel Approximation (GCA)

GCA assumes that the electric field perpendicular to the channel (transverse field E_x created by gate voltage) is much larger than the electric field parallel to the channel (longitudinal field E_y created by drain-to-source voltage): $|E_x| \gg |E_y| \Rightarrow \left| \frac{\partial E_x}{\partial x} \right| \gg \left| \frac{\partial E_y}{\partial y} \right|$.

n-channel MOSFET Cross-Section



1.1 Linear Region Derivation

Let the coordinate along the channel be y , where $y = 0$ at the source and $y = L$ at the drain. Let $V(y)$ be the channel potential at position y . The local mobile inversion charge density per unit area $Q_I(y)$ is:

$$Q_I(y) = -C_{ox} [V_{GS} - V(y) - V_{th}] \quad (1)$$

The drift current in the channel is driven by the longitudinal electric field $E_y = -dV(y)/dy$. The drain current I_D at any point y is:

$$I_D = W \cdot Q_I(y) \cdot v_y(y) \quad (2)$$

where W is the channel width, and $v_y(y)$ is the drift velocity of electrons: $v_y(y) = -\mu_n E_y(y) = \mu_n \frac{dV(y)}{dy}$. Substituting the expressions:

$$I_D = W \cdot (-C_{ox} [V_{GS} - V(y) - V_{th}]) \cdot \left(-\mu_n \frac{dV(y)}{dy} \right) \quad (3)$$

$$I_D = \mu_n C_{ox} W [V_{GS} - V(y) - V_{th}] \frac{dV(y)}{dy} \quad (4)$$

Rearranging terms:

$$I_D \cdot dy = \mu_n C_{ox} W [V_{GS} - V_{th} - V(y)] dV(y) \quad (5)$$

Integrating along the channel length from the source ($y = 0, V(0) = 0$) to the drain ($y = L, V(L) = V_{DS}$):

$$\int_0^L I_D dy = \mu_n C_{ox} W \int_0^{V_{DS}} [V_{GS} - V_{th} - V] dV \quad (6)$$

Since I_D is constant throughout the channel:

$$I_D \cdot L = \mu_n C_{ox} W \left[(V_{GS} - V_{th})V - \frac{V^2}{2} \right]_0^{V_{DS}} \quad (7)$$

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right] \quad (8)$$

This is the **MOSFET Linear Region Current Equation** (valid for $V_{DS} < V_{GS} - V_{th}$).

1.2 Saturation Region Derivation

As V_{DS} increases, the channel charge density at the drain end drops to zero when $V_{DS} = V_{GS} - V_{th}$. This condition is called **pinch-off**. The saturation current $I_{D,sat}$ remains constant with further increases in V_{DS} . Substituting the pinch-off voltage $V_{DS} = V_{GS} - V_{th}$ into the linear current equation:

$$I_{D,sat} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th})(V_{GS} - V_{th}) - \frac{(V_{GS} - V_{th})^2}{2} \right] \quad (9)$$

$$I_{D,sat} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 \quad (10)$$

This is the **MOSFET Saturation Region Current Equation** (valid for $V_{DS} \geq V_{GS} - V_{th}$).

2. MOSFET Scaling Principles

Scaling refers to reducing the physical dimensions of the MOSFET (channel length L , width W , gate oxide thickness t_{ox}) by a factor of $\beta > 1$ to increase packing density and speed.

2.1 Types of Scaling

- **Constant-Field Scaling (Full Scaling):** All physical dimensions (W, L, t_{ox}) and voltages (V_{DD}, V_{th}) are scaled down by the factor $1/\beta$. This keeps the internal electric fields constant, preventing breakdown.
- **Constant-Voltage Scaling:** Only physical dimensions are scaled down by $1/\beta$, while the supply voltage V_{DD} remains unchanged. This preserves system compatibility with older circuits (e.g. 5V logic) but leads to very high electric fields inside the chip.

2.2 Scaling Factors Summary

The table below summarizes how physical parameters scale under both scaling paradigms:

Comparison of Scaling Paradigms

Parameter	Constant-Field	Constant-Voltage
Dimensions (W, L, t_{ox})	$1/\beta$	$1/\beta$
Voltages (V_{DD}, V_{th})	$1/\beta$	1
Electric Field (E)	1	β
Capacitance (C_{ox})	β	β
Drain Current (I_D)	$1/\beta$	β
Gate Delay (τ)	$1/\beta$	$1/\beta^2$
Power Dissipation (P)	$1/\beta^2$	β
Power Density (P/Area)	1	β^3

> [!WARNING] > Under Constant-Voltage scaling, the power density increases by β^3 . This causes massive heat dissipation issues in sub-micron chips, making Constant-Field (or hybrid) scaling mandatory for modern deep sub-micron VLSI designs.

Quick Revision — Module 2

MOSFET Current Derivation & Scaling | PE-EC603C

Key Module 2 Formulas

$$\text{Linear Current: } I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$\text{Saturation Current: } I_{D,sat} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$$

$$\text{Pinch-off Condition: } V_{DS,sat} = V_{GS} - V_{th} \quad C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

One-Line Definitions

- **GCA:** Approximation assuming channel potential variations are much slower than gate oxide potential drops.
- **Pinch-off:** Condition where the inversion channel charge vanishes at the drain end ($V_{DS} = V_{GS} - V_{th}$).
- **Constant-Field Scaling:** Scaling method where dimensions and voltages are reduced by β to keep internal fields constant.
- **Constant-Voltage Scaling:** Scaling method where dimensions are reduced by β while voltages remain constant.
- **Subthreshold Current:** Leakage current that flows from drain to source even when $V_{GS} < V_{th}$.
- **Drift Velocity:** Average velocity attained by charged particles (electrons) in a material due to an electric field.

Frequently Asked / Viva Questions

Q1. State the Gradual Channel Approximation (GCA) and its physical validity.

GCA assumes the longitudinal field along the channel $E_y = dV/dy$ changes very slowly compared to the vertical field E_x across the oxide. It is highly valid for long-channel devices ($L \gg t_{ox}$), but breaks down in short-channel devices where E_y is extremely high.

Q2. Derive the drain current equation for a MOSFET operating in the linear region.

Starting with charge density $Q_I(y) = -C_{ox}(V_{GS} - V(y) - V_{th})$ and current $I_D = WQ_I(y)\mu_n(dV/dy)$, we integrate from $y = 0$ to L and $V = 0$ to V_{DS} to obtain the linear equation: $I_D = \mu_n C_{ox} \frac{W}{L} [(V_{GS} - V_{th})V_{DS} - V_{DS}^2/2]$.

Q3. Explain the concept of channel pinch-off.

Pinch-off occurs when the voltage difference between the gate and the channel at the drain end falls below the threshold voltage, i.e., $V_{GD} = V_{GS} - V_{DS} = V_{th}$. At

Frequently Asked / Viva Questions (contd.)

this point, the local inversion charge density at the drain becomes zero, and the current saturates.

Q4. Why does Constant-Voltage scaling lead to reliability issues in sub-micron devices?

Because the physical dimensions (L, t_{ox}) are scaled down by $1/\beta$ while supply voltage stays constant, the internal electric fields increase by a factor of β . This causes oxide breakdown, hot-carrier injection, and velocity saturation.

Q5. A MOSFET has $W = 10\ \mu\text{m}$, $L = 1\ \mu\text{m}$, $\mu_n C_{ox} = 100\ \mu\text{A}/\text{V}^2$, and $V_{th} = 0.8\ \text{V}$. Find I_D at $V_{GS} = 2.0\ \text{V}$ and $V_{DS} = 0.5\ \text{V}$.

Here $V_{GS} - V_{th} = 2.0 - 0.8 = 1.2\ \text{V}$. Since $V_{DS} = 0.5\ \text{V} < 1.2\ \text{V}$, the device is in the linear region.

$$I_D = 100 \times 10 [(1.2)(0.5) - 0.25/2] = 1000 [0.6 - 0.125] = 475\ \mu\text{A}.$$

Q6. Calculate the saturation current $I_{D,\text{sat}}$ for the same device ($V_{GS} = 2.0\ \text{V}$, $V_{DS} = 1.5\ \text{V}$).

Since $V_{DS} = 1.5\ \text{V} > 1.2\ \text{V}$, the device is in saturation.

$$I_{D,\text{sat}} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2 = \frac{1}{2} \cdot 100 \cdot 10 \cdot (1.2)^2 = 500 \cdot 1.44 = 720\ \mu\text{A}.$$

Q7. Why does power density remain constant under constant-field scaling?

Under constant-field scaling, the power P scales by $1/\beta^2$ and the chip area ($W \times L$) scales by $1/\beta^2$. Therefore, the ratio P/Area scales by $(1/\beta^2)/(1/\beta^2) = 1$, keeping the thermal density constant.

Q8. Derive the scaling factor of gate delay τ in constant-field scaling.

Gate delay is $\tau = \frac{C \cdot V}{I}$. Capacitance $C \propto W \cdot L/t_{ox}$ scales by $1/\beta$, voltage V scales by $1/\beta$, and current I scales by $1/\beta$. Thus, τ scales by $\frac{(1/\beta) \cdot (1/\beta)}{1/\beta} = 1/\beta$ (speed increases by β).

Q9. What is the effect of channel length modulation on the saturation drain current?

It adds a term $(1 + \lambda V_{DS})$ to the saturation current equation. This represents the fact that after pinch-off, the effective channel length shortens as V_{DS} increases, causing a small, linear increase in current rather than a flat saturation.

Q10. Why is electron mobility (μ_n) higher than hole mobility (μ_p)?

Electrons reside in the conduction band where they have lower effective mass and experience less lattice scattering than holes, which move through the valence band by bond-breaking. This makes $\mu_n \approx 2.5$ to 3 times higher than μ_p .